Geodesic pancyclicity and balanced 5-pancyclicity of 3-ary n-cube

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Abstract

Let G be a graph. For any two vertices x, $y \in V(G)$, a cycle C is called (x,y)-geodesic if there exists a shortest $x-y$ path of G lies on C; and a cycle C is called (x,y)-balanced if x, $y \in V(C)$ and the distance of x and y in C is equal to $\lfloor |V(C)|/2 \rfloor$ (i.e.,$d_C(x,y) = max\{d_C(u,v) \mid u,v \in V(C)\}$). A graph G is geodesic pancyclic if for any two vertices $x,y \in V(G)$, there exists a (x,y)-geodesic cycle of every length from $max\{2d(x,y),3\}$ to $|V(G)|$. A graph G is balanced k-pancyclic if for any two vertices $x,y \in V(G)$, there exists a (x,y)-balanced cycle of every length from $max\{2d(x,y),k\}$ to $|V(G)|$. In this paper, we shall show that the 3-ary n-cube Q_n^3 is geodesic pancyclicity and balanced 5-pancyclicity.

Keywords: geodesic pancyclic, balanced pancyclic, k-ary n-cube.

1 Introduction

Interconnection network is usually represented by a graph where each vertex represents a processor and each edge represents a communication link between two processors. A good interconnection network usually includes some beautiful properties, such as symmetry, smaller diameter, small degree, embedding capabilities, efficient routing, etc. Some structures of interconnection networks are popular graphs such as hypercubes, star graphs, k-ary n-cubes, and hypercube-like graphs.

Efficiency of data transfer is an important issue in the interconnection network. In order to improve the transmission latency, many literatures discussed the path and cycle embedding problems. In early period, the Hamiltonian and Hamiltonian-connected properties are studied (for example, see [15]). Later, many papers focused on the pancyclic (resp. panconnected) problems which try to find cycles (resp. paths) of various lengths in a graph [9, 10, 11]. A graph G is called *pancyclic* if it contain a cycle of every length l with $3 \leq l \leq |V(G)|$. Furthermore, G is called *vertex-pancyclic* (resp. *edge-pancyclic*) if each vertex (resp. edge) belongs to a cycle of every length l with $3 \leq l \leq |V(G)|$. Many hypercube-like graphs have been proved to be vertex-pancyclic and edge-pancyclic [7, 12]. When extending the role of a vertex/edge to a shortest path, Chan et al. [3] introduced the concept of geodesic pancyclic. For a graph G, if for any two vertices x and y, there exists a shortest $x-y$ path which is contained in a cycle of every (resp. even) length l with $max\{2d(x,y),3\} \leq l \leq |V(G)|$, G is called *geodesic pancyclic* (resp. *geodesic bipancyclic*). Chan et al. [3] gave some sufficient conditions of geodesic pancyclic graphs. The authors in [6, 8] proved that the augmented cubes and the crossed cubes are geodesic pancyclic.

Another variation of pancyclicity is the balanced pancyclicity. If for any two vertices x and y, there exist two internally disjoint $x-y$ paths P_1, P_2 with $|l(P_1) - l(P_2)| \leq 1$ whose union forms a cycle of every (resp. every even) length l with $max\{2d(x,y),k\} \leq l \leq |V(G)|$, G is called *balanced k-pancyclic* (resp. *balanced k-bipancyclic*). [6, 8] also proved that the augmented cubes are balanced pancyclic. Later, Tsai et al. [14] showed that the hypercube is geodesic bipancyclic and balanced bipancyclic.

The k-ary n-cube Q_n^k is a popular topology in interconnection networks. Due to its desirable properties, such as recursive structure, capability of efficiently simulating other topologies, ease of implementation, and ability to reduce message latency by exploiting communication locality found in many parallel applications [1, 13]. It has been investigated extensively.

About the path and cycle embedding properties on k-ary n-cubes, Wang et al. [15] showed that the k-ary n-cube is almost Hamiltonian-connected, and Hamiltonian-connected if k is odd. Hsieh et al. [5] showed that the 3-ary n-cube is panconnected and edge-pancyclic. Moreover, Hsieh and Lin [4] proved that Q_n^k is both $\lfloor k/2 \rfloor \times n$-panconnected and k-edge-pancyclic when $k \geq 3$ is odd, and Q_n^k is both bipanconnected and edge-bipancyclic when k is even. In this paper, we will prove that the 3-ary n-cube is geodesic pancyclic and balanced 5-pancyclic.

The rest of this paper is organized as follows: necessary definitions and properties of k-ary n-cube is introduced in Section 2. In Section 3, we will prove that the 3-ary n-cube is geodesic pancyclic. In Section 4, we shall show that the 3-ary n-cube is balanced 5-pancyclic. Finally, some concluding remarks are given in Section 5.

2 Preliminaries

A simple graph $G = (V, E)$ is a graph with no loops or multiple edges, where V is the vertex set and E is the edge set. Two vertices u and v are *adjacent* if (u, v) is an edge of G. A *subgraph* of $G = (V, E)$ is a graph $G' = (V', E')$ such that $V' \subseteq V$ and $E' \subseteq E$. An *induced subgraph* is an edge-preserving subgraph, that is, G' is an induced subgraph of G iff $V' \subseteq V$ and $E' = \{(x, y) \in E | x, y \in V'\}$.

A *path* is a sequence of adjacent vertices, written as $P = <v_0, v_1, v_2, \ldots, v_m>$, where v_0 is called the *source-vertex* and v_m is called the *sink-vertex* of the path. P is also called a $v_0 - v_m$ *path*. The *length* of a path P, denoted by $l(P)$, is the number of edges in P. A $v_0 - v_m$ path P may contain a subpath P', denoted by $P = <v_0, v_1, \ldots, v_j, P', v_k, v_{k+1}, \ldots, v_m>$ where $P' = <v_j, v_{j+1}, \ldots, v_{k-1}, v_k>$. A *cycle* is a path with at least three vertices such that the source-vertex is the same as the sink-vertex, written as $C = <v_0, v_1, v_2, \ldots, v_m, v_0>$. A l-cycle is a cycle of length l. Give a subgraph H of G, the *distance between u and v* in H, denoted by $d_H(u, v)$, is the length of a shortest $u - v$ path in H. If $H = G$, we write $d(u, v)$ for short. The *diameter* of G is the maximum distance between any pair of vertices of G. A cycle (resp. path) is called a *Hamiltonian cycle* (resp. *Hamiltonian path*) if it traverses every vertex of G exactly once.

A graph G is *edge-pancyclic* if for every edge e, G contains a cycle C of every length l from 3 to $|V(G)|$ such that e is in C. G is called *panconnected* if any two distinct vertices u, v are joined by a path of every length l from $d(u, v)$ to $|V(G)|$. Sometimes, the lower bound $d(u, v)$ of l is difficult to achieved. If any two vertices u, v can be connected by a path of every length l ranging from k to $|V(G)|$, we say that G is *k-panconnected.*

The *k-ary n-cube* Q_n^k ($k \geq 2$ and $n \geq 1$) has k^n vertices each of the form $x = x_n x_{n-1} \ldots x_1$, where $0 \leq x_i < k$ for all $1 \leq i \leq n$. Two vertices $x = x_n x_{n-1} \ldots x_1$ and $y = y_n y_{n-1} \ldots y_1$ in Q_n^k are adjacent if and only if there exists an integer j with $1 \leq j \leq n$, such that $x_j = y_j \pm 1 (\text{mod } k)$ and $x_l = y_l$ for every $l \in \{1, 2, \ldots, n\} - \{j\}$. Obviously, Q_1^k is a k-cycle, Q_n^2 is the n-dimensional hypercube, and Q_2^k is a $k \times k$ wrap-around mesh. Figure 1 illustrates Q_1^5, Q_2^3, and Q_3^3. It is shown in [2] that each vertex of Q_n^k has degree $2n$ when $k \geq 3$ and degree n when $k = 2$. Moreover, the diameter of Q_n^k is $n \times \lfloor k/2 \rfloor$.

The ith position, from the right to the left, of the n-bit string $x_n x_{n-1} \ldots x_1$ is called the *i-dimension*. We can partition Q_n^k along any i-dimension by regarding the graph comprised by k disjoint copies $Q_{n-1}^k[0]$, $Q_{n-1}^k[1]$, $Q_{n-1}^k[2]$, $\ldots$, and $Q_{n-1}^k[k-1]$, where $Q_{n-1}^k[j]$ is the subgraph of Q_n^k induced by $\{x \in V(Q_n^k)|$ the ith bit x_i of $x = x_n x_{n-1} \ldots x_1$ is fixed by $j\}$. Each $Q_{n-1}^k[j]$ is called a *subcube* of Q_n^k. Note that each $Q_{n-1}^k[j]$ is isomorphic to a k-ary $(n-1)$-cube. Clearly, the k^{n-1} edges between $Q_{n-1}^k[j]$ and $Q_{n-1}^k[j+1]$ form a perfect matching, where $j+1$ is taken to module k. So $Q_{n-1}^k[j]$ and $Q_{n-1}^k[j+1]$ are called *adjacent*

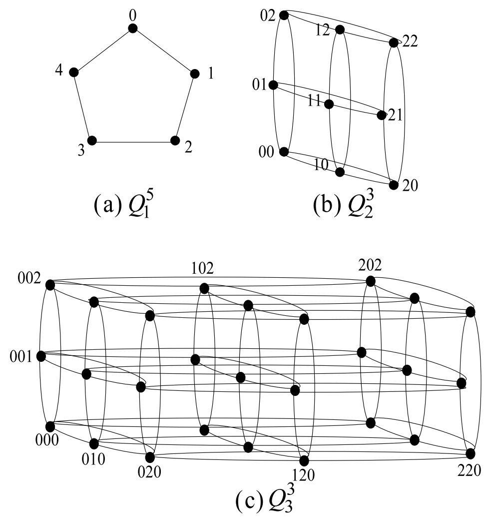

Figure 1: Examples of some k-ary n-cube.

subcubes, and the edges between them are called *crossing edges*. We use the notation "$u \rightarrow v$" to indicate that (u, v) is a crossing edge between two adjacent subcubes.

Definition 1 Let G be a graph. For any two vertices $x, y \in V(G)$, a cycle is called (x, y)-*geodesic* if there is a shortest $x - y$ path which lies on the cycle. A (x, y)-*geodesic* l-*cycle* in G is a (x, y)-geodesic cycle of length l. G is called *geodesic pancyclic* if for any two vertices $x, y \in V(G)$, and every integer l satisfying $\max\{2d(x, y), 3\} \leq l \leq |V(G)|$, a (x, y)-geodesic l-cycle exists.

Definition 2 Let G be a graph. For any two vertices $x, y \in V(G)$, a cycle C is called (x, y)-*balanced* if $x, y \in V(C)$ and $d_C(x, y) = \max\{d_C(u, v) | u, v \in V(C)\} = \lfloor |V(C)|/2 \rfloor$. A (x, y)-*balanced* l-*cycle* in G is a (x, y)-balanced cycle of length l. G is called *balanced* k-*pancyclic* if for any two vertices $x, y \in V(G)$, and every integer l satisfying $max\{2d(x, y), k\} \leq l \leq |V(G)|$, a balanced l-cycle exists.

Lemma 1 [2] The diameter of Q_n^k is $\lfloor k/2 \rfloor \times n$.

Lemma 2 [5] The graph Q_n^3 is n-panconnected and edge-pancyclic.

The following lemmas are two useful observations of Q_n^3.

Lemma 3 Let G_0, G_1 be two adjacent subcubes of Q_n^3. If $x \in V(G_0)$, $y \in V(G_1)$ and $d(x, y) \geq 2$, then there exist two shortest $x - y$ paths P_0 and P_1 such that all vertices of P_0 are in G_0 except y, and all vertices of P_1 are in G_1 except x.

Proof: Let x' be the vertex in G_1 such that $x \rightarrow x'$. let y' be the vertex in G_0 such that $y \rightarrow y'$. If $d(x, y) = d$, it follows that the n-bit strings of x and y have exactly d different bits. Since $x, y' \in V(G_0)$ and $x', y \in V(G_1)$, it is clear that $d(x, y') = d - 1$ and $d(x', y) = d - 1$. Hence, there exists a shortest $x - y'$ path P_A in G_0 of length $d - 1$, and a shortest $x' - y$ path P_B in G_1 of length $d - 1$. Let $P_0 = <x, P_A, y', y>$ and $P_1 = <x, x', P_B, y>$. It follows that they are shortest $x - y$ paths such that all vertices of P_0 are in G_0 except y, and all vertices of P_1 are in G_1 except x. □

Lemma 4 Let G_0, G_1 be two adjacent subcubes of Q_n^3, and C be a cycle (resp. path) of length l in $Q_n^3 - G_1$ such that $|E(C \cap G_0)| \geq 1$. If $e \in E(C \cap G_0)$, then there exists a cycle (resp. path) C' in Q_n^3 containing $C - e$, and $l' = l(C')$ is any number with $l + 2 \leq l' \leq l + 3^{n-1}$. We say that C can be extend to C' by the edge e.

Proof: Let $e = (x, y)$, and let $x', y' \in V(G_1)$ such that $x \rightarrow x'$ and $y \rightarrow y'$. It clear that $(x', y') \in E(G_1)$. Since G_1 is edge-pancyclic by Lemma 2, there exists an $x' - y'$ path P in G_1 of length $l' - l - 1$. It follows that $C' = C \cup P - e$ is a cycle (or path) of length l'. □

3 Geodesic pancyclicity of 3-ary n-cube

In this section, we will prove that the 3-ary n-cube is geodesic pancyclic.

Lemma 5 The 3-ary 2-cube Q_2^3 is geodesic pancyclic.

Proof: Choose any two vertices $x \neq y \in V(Q_2^3)$. Since the diameter of Q_2^3 is 2 by Lemma 1. The distance of x and y must be 1 or 2.

Case 1: $d(x, y) = 1$.
By Lemma 2, Q_2^3 is edge-pancyclic. There exists a cycle containing the edge (x, y) of every length from 3 to 8.

Case 2: $d(x, y) = 2$.
Without loss of generality, we assume that $x = 00$ and $y = 11$. It is clear that $P = <00, 10, 11>$ is a shortest $x - y$ path. For each l with $4 \leq l \leq 9$, Figure 2 shows a (x, y)-geodesic l-cycle containing P. □

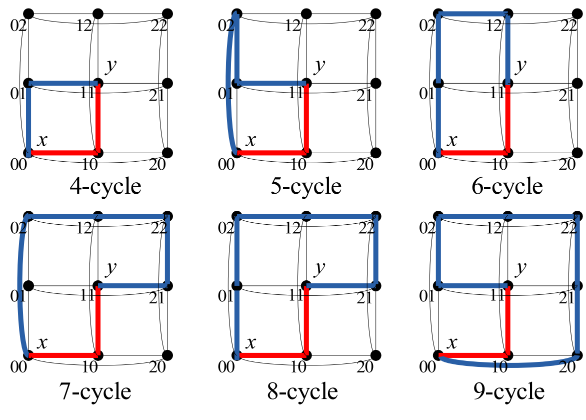

Figure 2: (x, y)-geodesic cycles for $x = 00$ and $y = 11$.

Theorem 1 For any two vertices $x, y \in V(Q_n^3)$, and any integer l satisfying $max\{2d(x, y), 3\} \leq l \leq 3^n$, there exists a (x, y)-geodesic l-cycle.

Proof: We prove this theorem by induction on n. The result holds for $n = 1$, because Q_1^3 is 3-cycle. The result for $n = 2$ also holds by Lemma 5. Suppose that this result holds for Q_{n-1}^3, now we consider Q_n^3 for $n \geq 3$. Let x and y be two distinct vertices of Q_n^3. Since Q_n^3 is vertex-transitive, we can assume that $x = 00\ldots00$. There are the following two cases.

Case 1: $d(x, y) < n$.
Because $d(x, y) < n$, there exists some ith bit of y is 0, we can partition Q_n^3 along the dimension i into three subcubes G_0, G_1, G_2, where $G_j = Q_{n-1}^3[j]$ for $j = 0, 1, 2$. Hence x and y are in the same subcube G_0.

Subcase 1.1: $max\{2d(x, y), 3\} \leq l \leq 3^{n-1}$.
By the induction hypothesis, G_0 is geodesic pancyclic. Hence, there exists a (x, y)-geodesic l-cycle.

Subcase 1.2: $3^{n-1} + 1 \leq l \leq 2 \times 3^{n-1}$.
Choose a (x, y)-geodesic cycle C of length $3^{n-1} - 1$ (or $3^n - 1$) in the subcube G_0. Let P be the shortest $x - y$ path lying on C. By Lemma 4, the cycle C can be extend to get a (x, y)-geodesic l-cycle in $G_0 \cup G_1$ by any edge $e \in E(C - P)$.

Subcase 1.3: $2 \times 3^{n-1} + 1 \leq l \leq 3^n$.
Similarly, by Subcase 1.2, there is a (x, y)-geodesic cycle C of length $2 \times 3^{n-1} - 1$ (or $2 \times 3^{n-1}$) in the subgraph $G_0 \cup G_1$. Let P be the shortest $x - y$ path lying on C. It is clear that $P \subseteq G_0$. By Lemma 4, the cycle C can be extend to a (x, y)-geodesic l-cycle by any edge $e \in E(C \cap G_1)$. See Figure 3 for illustration.

Case 2: $d(x, y) = n$.
Without loss of generality, assume that some ith bit of y is 1. Partition Q_n^3 along the dimension i into three subcubes G_0, G_1, G_2, where $G_j = Q_{n-1}^3[j]$ for $j = 0, 1, 2$. Hence $y \in V(G_1)$. We have the following two subcases.

Subcase 2.1: $2n \leq l \leq 2 \times 3^{n-1}$.
Let x_1 be the vertex of G_1 such that $x \to x_1$. Note that $x_1 \neq y$ by $d(x, y) = n$. Since G_1 is geodesic pancyclic by the induction hypothesis, there exists a (x_1, y)-geodesic cycle C of every length l_1 with $2 \times (n - 1) \leq l_1 \leq 3^{n-1}$, where C=$<x_1, P_s, y, P_c, x_1>$, and P_s is a shortest $x_1 - y$ path. Note that $l(P_s) = n - 1$ and $l(P_c) = l_1 - n + 1 \geq n - 1$. Let z be the vertex of P_c with $(x_1, z) \in E(P_c)$. By Lemma 4, we can extend the cycle C by the edge (x_1, z) to get a (x, y)-geodesic cycle of length l. See Figure 4(a) for illustration.

Subcase 2.2: $2 \times 3^{n-1} + 1 \leq l \leq 3^n$.
By Subcase 2.1, there is a (x, y)-geodesic cycle C of length $2 \times 3^{n-1} - 1$ (or $2 \times 3^{n-1}$) in the subgraph $G_0 \cup G_1$ with a shortest $x - y$ path P_s. By Lemma 4, the cycle C can be extend to a (x, y)-geodesic l-cycle by any edge $e \in E(C \cap G_1 - P_s)$. See Figure 4(b). □

Corollary 1 The 3-ary n-cube Q_n^3 is geodesic pancyclic.

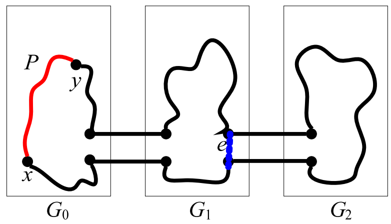

Figure 3: An illustration of Subcase 1.3 in the proof of Theorem 1.

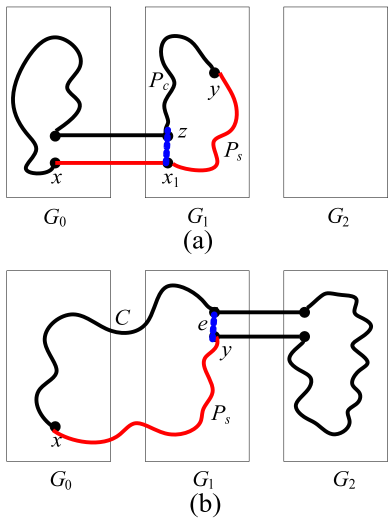

Figure 4: An illustration of Case 2 in the proof of Theorem 1.

4 Balanced 5-pancyclicity of 3-ary n-cube

In this section, we study the balanced pancyclicity of Q_n^3. Since there is no (x, y)-balanced 4-cycle in the graph Q_2^3 when x and y are adjacent vertices, we only discuss the balanced 5-pancyclicity of Q_n^3.

Lemma 6 The 3-ary 2-cube Q_2^3 is balanced 5-pancyclic.

Proof: Choose any two vertices x and y of Q_2^3. Due to the vertex-transitivity, edge-transitivity of Q_2^3, and the diameter of Q_2^3 is 2 by Lemma 1, we need only to consider the following two cases.

Case 1: $d(x, y) = 1$.
Without loss of generality, assume that $x = 00$ and $y = 01$. Figure 5 illustrates those (x, y)-balanced cycles of all lengths ranging from 5 to 9.

Case 2: $d(x, y) = 2$.
Without loss of generality, assume that $x = 00$ and $y = 11$. Figure 6 illustrates those (x, y)-balanced cycles of all lengths ranging from 5 to 9. □

Theorem 2 For any two vertices $x, y \in V(Q_n^3)$, and any integer l satisfying $max\{2d(x, y), 5\} \leq l \leq 3^n$, there exists a (x, y)-balanced l-cycle.

Proof: We prove this theorem by induction on n. The result for $n = 1$ holds trivially, By Lemma 6, the result holds for $n = 2$. Assuming that this result is true for Q_{n-1}^3, now we consider Q_n^3 for $n \geq 3$. Let x and y be two distinct vertices of Q_n^3. Since Q_n^3 is vertex-transitive, we can assume that $x = 00\ldots0$. There are the following two cases.

Case 1: $d(x, y) < n$.
Because $d(x, y) < n$, there exists some ith bit of y is 0, we can partition Q_n^3 along the dimension i into three subcubes G_0, G_1, G_2, where $G_j = Q_{n-1}^3[j]$ for $j = 0, 1, 2$. Hence x and y are in the same subcube G_0.

Subcase 1.1: $max\{2d(x, y), 5\} \leq l \leq 3^{n-1}$.
By the induction hypothesis, G_0 is balanced 5-pancyclic. Hence, there exists a (x, y)-balanced l-cycle.

Subcase 1.2: $3^{n-1} + 1 \leq l \leq 2 \times 3^n$.
Write $l = l_0 + 2 \times l_1$, where $3^{n-1} - 3 \leq l_0 \leq 3^{n-1}$ and $2 \leq l_1 \leq 3^{n-1}$. Let $C = <x, P_1, y, P_2, x>$ be a (x, y)-balanced cycle in G_0 of length l_0. By Lemma 4, since G_0, G_1 are adjacent subcubes, we can extend P_1 by any edge $e_1 \in E(P_1)$ to a longer path P_1' in $G_0 \cup G_1$ with length $l(P_1') = l(P_1) + l_1$. Similarly, since G_0, G_2 are adjacent subcubes, P_2 can be extended by any edge $e_2 \in E(P_2)$ to a longer path P_2' in $G_0 \cup G_2$ with length $l(P_2') = l(P_2) + l_1$. Therefore, $P_1' \cup P_2'$ forms a (x, y)-balanced cycle of length l. See Figure 7 for illustration.

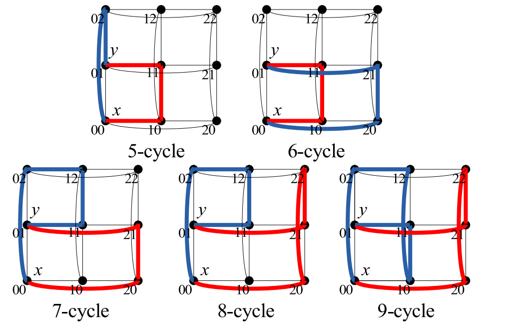

Figure 5: (x, y)-balanced cycles for $x = 00$ and $y = 01$.

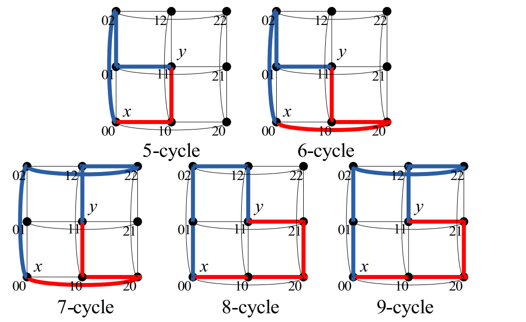

Figure 6: (x, y)-balanced cycles for $x = 00$ and $y = 11$.

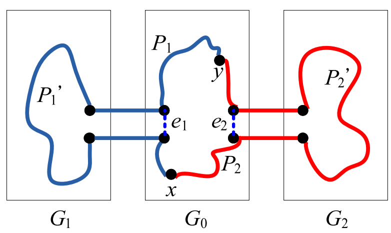

Figure 7: An illustration of Case 1 in the proof of Theorem 2.

Case 2: $d(x, y) = n$.

Without loss of generality, assume that some ith bit of y is 1. Partition Q_n^3 along the dimension i into three subcubes G_0, G_1, G_2, where $G_j = Q_{n-1}^3[j]$ for $j = 0, 1, 2$. Hence $y \in V(G_1)$. We have the following two subcases.

Subcase 2.1: $2n \leq l \leq 2 \times 3^{n-1}$.

Write $l = l_0 + l_1 + 2$, where $n - 1 \leq l_0 \leq l_1 \leq 3^{n-1} - 1$, and $l_1 - l_0 \leq 1$. Let x' be the vertexof $V(G_1)$ such that and $x \to x'$, and y' be the vertex of $V(G_0)$ such that and $y \to y'$. By Lemma 2, G_0 and G_1 are $(n-1)$-panconnected. Hence, there exists a $x - y'$ path P_0 in G_0 of length l_0, and a $x' - y$ path P_1 in G_1 of length l_1. Hence $P_0 \cup P_1 \cup \{(x, x'), (y, y')\}$ is a (x, y)-balanced cycle of length l. See Figure 8(a) for illustration.

Subcase 2.2: $2 \times 3^{n-1} + 1 \leq l \leq 3^n$.

Write $l = l_0 + 2 \times l_1$, where $3^{n-1} - 3 \leq l_0 \leq 3^{n-1}$and $2 \in l_1 \in 3^{n-1}$. Let x_2 and y_2 be vertices in G_2 such that $x \to x_2$ and $y \to y_2$. Note that $x_2 \neq y_2$ since $d(x, y) = n \geq 3$. Hence there is a (x_2, y_2)-balanced cycle C of length l_0 by the induction hypothesis. Let C=$<x_2, P_1, y_2, P_2, x_2>$, and let x_2' be the vertex of P_1 with $e_1 = (x_2, x_2') \in E(P_1)$. By Lemma 4, since G_0, G_2 are adjacent subcubes, we can extend P_1 by e_1 to a longer path P_1' in $G_0 \cup G_2$with length $l(P_1') = l(P_1) + l_1$. Similarly, let y_2' be the vertex of P_2 with $e_2 = (y_2, y_2') \in E(P_2)$. By Lemma 4, since G_1, G_2 are adjacent subcubes, P_2 can be extend by e_2 to a longer path P_2' in $G_1 \cup G_2$ with length $l(P_2') = l(P_2) + l_1$. Therefore, $P_1' \cup P_2'$ forms a (x, y)-balanced cycle of length l. (See Figure 8 (b).) □

Corollary 2 The 3-ary n-cube Q_n^3 is balanced 5-pancyclic.

5 Conclusions

A geodesic cycle can minimize the transmission delay from x to y, and a balanced cycle can improve the efficiency of data transfer. In this paper, we showed that the 3-ary n-cube is geodesic pancyclic and balanced 5-pancyclic. A future work will extend these results to discuss the k-ary n-cube for $k \geq 3$.

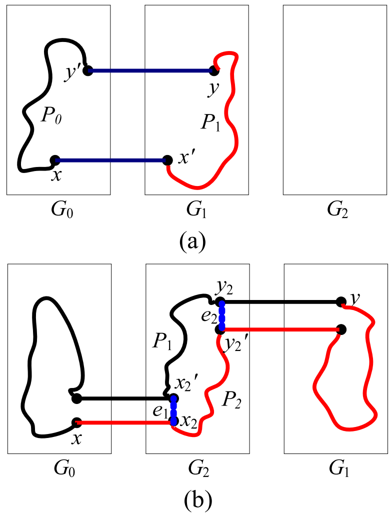

Figure 8: Illustrations of Case 2 in the proof of Theorem 2.

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